

CHAPTER 1

INTRODUCTION

Generative artificial intelligence has become part of how digital content is created. Image-generation tools can now produce detailed pictures from short text prompts, making them useful in design, education, communication, and other creative work. At the same time, the same technology can make it harder to tell whether a digital image came from a real source or was generated using AI. This creates concerns when generated content is presented as genuine, especially in online materials where people may only see the image and have little information about where it came from. Research on Generative AI has also pointed to concerns involving authenticity, misinformation, and responsible use of generated content (Nah et al., 2023).

This study focuses on a smaller part of that problem by examining images that contain visible text. Instead of using the meaning of the words, the study will look at how the text visually appears in the image and whether measurable features from those text regions can provide useful information for classifying Non-AI-Generated images and AI-generated images. The study combines text-region detection, visual feature extraction, and traditional machine-learning classification in a reproducible experimental process. This chapter presents the Project Context or Background of the Study, Statement of the Problem, Objectives of the Study, Significance of the Research, Scope and Limitations of the Project, and Definition of Terms.



Generative AI has changed the way digital images can be created. Modern image-generation tools can produce photographs, illustrations, posters, and other visual materials from short prompts or reference images. This gives people faster ways to create content, but it also creates a practical problem: generated images can be presented without clear information about how they were made. In some cases, they may be used in misleading posts, false announcements, fabricated document-like materials, or other content that appears genuine at first look. For this reason, distinguishing AI-generated images from real images has become an active problem in computer vision and digital image analysis.

The problem is not simply that AI-generated images exist. The larger difficulty is that their quality continues to improve, while image generators do not all produce the same visual patterns. A detector that performs well on images produced by one generator may not perform the same way when it is tested on images produced by another. Zhu et al. (2023) introduced GenImage, a large benchmark containing real and AI-generated images from several modern generators. Their evaluation included cross-generator testing and degraded-image testing, showing the importance of checking whether a detection method can still perform when the image source or image quality changes.

A similar concern was examined by Ojha et al. (2023). Their study showed that detectors trained using images from one type of generative model may fail when they encounter images from newer or unseen model families. This finding is important because it means that a high result on one dataset does not



automatically make a detector reliable for every AI-generated image. More recently, Yan et al. (2025) tested several existing detectors on a challenging dataset of realistic AI-generated images. Most of the tested methods frequently classified the generated images as real. Their work also combined different forms of visual evidence, including high-level representations and low-level patch information. Together, these studies show that AI-generated image detection is still affected by the dataset, generator, and type of visual information being examined.

Existing detection methods use different kinds of evidence. Some methods learn complex patterns from the complete image, while others focus on visual artifacts, frequency information, noise, or smaller local regions. Sinitsa and Fried (2024), for example, studied fingerprints left by generative models and showed that generated images can contain visual traces that are useful for detection. Zhong et al. (2023) also explored a localized approach through PatchCraft, where texture patches were examined instead of depending mainly on the global semantic content of the image. Their results support the broader idea that useful information may sometimes be found in specific image regions rather than only in the image as a whole.

This idea becomes relevant when the image contains a large amount of visible text. Text is visually different from many other image regions because it contains repeated strokes, edges, character shapes, spacing patterns, and alignment structures. These properties can create dense visual detail, especially in posters, announcements, social-media graphics, document-like images, and



other text-rich materials. However, text-rich images are not automatically easier to classify. Zhang et al. (2026) introduced TextFake, a benchmark designed specifically for AI-generated image detection on text-rich images. Their evaluation reported a clear performance gap between ordinary natural-image benchmarks and text-rich images and described a 'Text Density Curse,' where increasing text density can make detection more difficult for some methods.

The findings from TextFake are important to the present study because they show that visible text can change the behavior of AI-image detectors. Dense letters and glyph edges may interfere with the low-level patterns that a detector normally uses. At the same time, this raises another question that is not fully answered by general whole-image detection: instead of treating text mainly as something that makes detection harder, can the visual appearance of the text regions themselves provide useful classification information? That question is different from reading the words in the image. It is concerned with how the text looks as visual material.

Text-region detection provides a practical way to isolate these parts of an image. A text detector or OCR tool can locate areas that contain visible text, allowing the study to crop or process those areas separately. In this research, OCR may be used as a locating tool, but the recognized words will not be used as classification features. The spelling, language, topic, and meaning of the text are outside the intended feature set. This restriction is important because the study is not trying to determine whether certain words are more common in



AI-generated images. It is testing whether measurable visual properties inside the text regions carry useful information.

After the text regions are identified, selected visual measurements can be extracted from them. Candidate feature groups in the study include text density, stroke appearance, character shape, spacing, alignment, edge quality, texture, contrast, and sharpness. The exact numerical definitions will be finalized during the methodology and applied consistently across the dataset. If one image contains several detected text regions, their measurements will be combined into a consistent image-level feature record. These records can then be used by traditional machine-learning classifiers such as Logistic Regression, Support Vector Machine, and Random Forest. The goal of using these classifiers is not to claim that one of them is a universal AI-image detector, but to test whether the selected visual text-region features contain enough information to separate the two classes under controlled conditions.

The research gap therefore lies in the limited focused evaluation of visual features taken specifically from text regions for non-AI-generated-image -versus-AI-generated image classification without using the recognized text or its meaning. Existing research has already shown that AI-generated image detection can fail across generators and datasets, that generated images may contain measurable visual traces, that local image regions can be useful for analysis, and that text-rich images create a difficult detection setting. What remains less clear is whether selected visual characteristics from the text regions



themselves can provide useful, limited, or inconsistent evidence for the classification task.

To address this gap, the study will develop and evaluate a reproducible machine-learning experimental pipeline. The process will prepare selected Non-AI-generated text-containing images and AI-generated text-containing images, detect their visible text regions, extract selected visual features, organize those measurements into image-level records, and compare selected traditional machine-learning classifiers. The study will also examine the contribution of feature groups and document the limitations observed during testing. The exact dataset category and data sources will remain tentative until the ongoing dataset research and pilot testing are completed. Accordingly, the conclusions will be limited to the final selected dataset and controlled experimental conditions. A simple prototype may be developed to demonstrate the process, but the primary contribution of the study is the documented and reproducible experimental pipeline rather than a production-ready authenticity detector.

STATEMENT OF THE PROBLEM

The goal of this study is to determine if visual features extracted from regions of text are useful for classifying AI-generated and non-AI-generated images containing text. Specifically, it seeks to answer the following questions:

1. What visual features can be extracted from the text regions of selected text-containing images?



2. How accurately can selected machine-learning classifiers distinguish non-AI-generated images from AI-generated images using these features?
3. Which groups of visual text-region features provide useful information for the classification task?
4. What limitations can be observed when the approach is tested on the selected dataset?

OBJECTIVES OF THE STUDY

The main objective of this study is to develop and evaluate a reproducible machine-learning classification approach for determining whether selected visual text-region features provide useful information for distinguishing Non-AI-generated text-containing images and AI-generated text-containing images. The study will focus on preparing the data, extracting visual information from detected text regions, comparing selected traditional classifiers, and examining the usefulness and limitations of the feature groups. The study specifically aims to address the following objectives:

1. To identify and prepare a suitable dataset of Non-AI-generated and AI-generated text-containing images, to ensure that the collected images are properly selected and suitable for the classification experiment;
2. To identify the visible text regions and extract selected visual text-region features related to stroke appearance, character shape, spacing, alignment, edge quality, texture, contrast, sharpness, and text density;
3. To implement and compare selected traditional machine-learning classifiers using the extracted visual text-region features and evaluate their



performance using accuracy, precision, recall, F1-score, and confusion matrix; and

4. To examine the usefulness of the selected visual text-region feature groups and determine the observed strengths and limitations of the approach under the selected dataset and testing conditions.

SIGNIFICANCE OF THE RESEARCH

This study may contribute to a clearer understanding of how the visual appearance of text regions can be used in the classification of Non-AI-generated text-containing images and AI-generated text-containing images. Its value is mainly in the findings, the documented feature groups, and the reproducible experimental process rather than in presenting a final tool for proving image authenticity. The following groups may benefit from the study:

Content Reviewers. Content reviewers may use the findings as additional reference when considering the possible strengths and limits of automated classification for text-containing images. The study may help them understand that a model prediction is only one form of evidence and should not replace source checking, fact-checking, or other verification procedures.

Organizations Handling Digital Materials. Organizations that work with online posts, announcements, document-like images, or other digital materials may gain a clearer understanding of the challenges involved in distinguishing real and AI-generated visual content. The study may also provide a technical reference when they assess whether automated image-checking methods are appropriate for a particular use.



Digital-Media Researchers. Researchers working on digital media, synthetic-image detection, image forensics, or related areas may use the results as evidence on whether visual text-region features provide useful, limited, or inconsistent information. The documented pipeline may also help them compare this region-focused approach with whole-image or deep-learning methods.

Future Developers. Future developers of AI-generated content detection systems may use the findings of this study as a reference when improving existing detection methods. The study may help them understand how text-region features can be added as another layer of analysis together with other detection approaches. It may also provide ideas on whether visual features from text areas can help improve the accuracy and reliability of future AI-content detection systems.

Computer Science Students. Students may use the study as a practical example of how computer vision, feature extraction, and traditional machine learning can be connected in one experimental process. It may also show how a research question can be tested without developing a new deep-learning architecture.

Researchers. The researchers may gain experience in dataset preparation, text-region detection, visual feature design, machine-learning classification, controlled evaluation, and interpretation of experimental limitations. The study will also provide the group with evidence that can be used to refine the final methodology after dataset research and pilot testing.



Future Researchers. Future researchers may use the documented procedures, feature groups, results, and limitations as a starting point for related work. They may test other datasets, image categories, generators, feature definitions, classifiers, languages, or robustness conditions while comparing their findings with the controlled results of this study.

SCOPE AND LIMITATIONS OF THE PROJECT

The study will focus on selected digital images that contain clearly visible text. These may include text-rich materials such as receipts, posters, announcements, social-media graphics, and similar visual content, depending on data availability. The exact image category, dataset source, number of samples, AI generators represented, and inclusion criteria will be finalized after the ongoing dataset research and pilot testing. The dataset will contain two classes: Non-AI-generated and AI-generated. Images will refer to the selected non-AI-generated images obtained from appropriate sources and AI-generated images will refer to images produced by selected generative models or identified as AI-generated in the chosen public dataset.

Both classes will be prepared using consistent experimental procedures. File format, image dimensions, resolution, compression, and other source-related properties will be checked and controlled where practical so that obvious differences between data sources do not become unintended classification signals. The prepared data will be divided into separate training, validation, and test sets. The final test set will be reserved for final evaluation and will not be used when choosing features, classifiers, or model settings.



The algorithm and feature component of the study will focus only on visual information extracted from detected text regions. Candidate feature groups include text density, stroke appearance, character shape, spacing, alignment, edge quality, texture, contrast, and sharpness. The exact numerical definitions of these features will be established in the methodology and applied consistently. Text detection or OCR may be used to locate visible text regions. If the selected tool also produces recognized text, the words, spelling, language content, topic, and meaning will not be used as classification features.

Selected traditional machine-learning classifiers will be implemented and compared. Candidate classifiers may include Logistic Regression, Support Vector Machine, and Random Forest, although the final set will be established based on feasibility and the results of methodology development. The study will not develop a new deep-learning architecture. Its purpose is to evaluate the selected visual text-region features using established classification methods.

The main technical output of the study will be a reproducible experimental pipeline covering image preparation, text-region detection, visual feature extraction, creation of image-level feature records, classifier training, and evaluation. When an image contains multiple text regions, the measurements from those regions will be combined using one consistent image-level representation. The exact aggregation method will be finalized in the methodology. A simple prototype may also be developed to demonstrate image input, text-region processing, feature extraction, prediction, and result display.



The prototype will not be treated as the main research contribution and will not be presented as a public, forensic, or production-ready detector.

The selected classifiers will be evaluated using accuracy, precision, recall, F1-score, and confusion matrix under the same prepared data and testing conditions. The study will also conduct controlled comparisons of the selected feature groups to identify which types of visual information appear useful, limited, or inconsistent for the classification task. If a prototype is developed, only basic functional testing will be performed to confirm that its demonstration steps operate as intended.

The study is limited by the size, quality, labels, and composition of the selected dataset. It is also dependent on the text-region detection method because missed, incorrect, or differently divided text regions may affect the extracted measurements. The visual features may be influenced by resolution, resizing, compression, font style, text size, and other image properties even when consistent preparation is applied. The study is limited to binary classification and will not identify the exact image generator, prove image provenance, determine legal authenticity, or replace professional verification. Unseen-generator testing, broad cross-generator evaluation, multilingual comparison, and image-degradation experiments such as blur, resizing, and JPEG compression are outside the initial scope. Therefore, the findings will apply only to the selected dataset and controlled testing conditions.



DEFINITION OF TERMS

The following terms are operationally defined according to how they are used in this study:

AI-Generated Image - An image included in the study that was produced by a selected generative AI image model or labeled as AI-generated by the selected public dataset.

Classification - The task in this study of assigning an image to one of two classes, real or AI-generated, using the selected machine-learning classifier and extracted visual text-region features.

Feature Extraction - The process of measuring selected visual properties from detected text regions and converting them into numerical values that can be used by the classifiers.

F1-Score - A classification performance measure used in this study to summarize the balance between precision and recall.

Image-Level Feature Record - The consistent numerical representation created for one image after the measurements from its detected text region or regions are organized or combined.

Machine Learning - The computational approach used in this study to train selected traditional classifiers using the extracted visual text-region feature records.

Real Image - A selected non-AI-generated image used as the real class in the study and obtained from an appropriate dataset or source under the study's final data rules.

Reproducible Experimental Pipeline - The documented sequence of procedures used in the study for data preparation, text-region detection, visual feature extraction, image-level feature preparation, classifier training, and evaluation so that the experiment can be repeated using the same or comparable data.

Text-Containing Image - A digital image included in the study that contains visible text regions that can be detected and analyzed.

Text Region - A detected area of an image that contains visible text and is processed separately for visual feature extraction.

Text-Region Detection - The process used to locate the parts of an image that contain visible text. OCR or another text-detection tool may be used for this purpose, but recognized words are not used as classification features.

Visual Text-Region Features - The numerical visual measurements extracted from detected text regions in this study, which may include properties related to text density, stroke appearance, character shape, spacing, alignment, edge quality, texture, contrast, and sharpness.

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